Track 22: Arduino

June 2026

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MR ROKESH TECHNOLOGY — Training & Internship Programs

> ***Skill path:*** *Electronics → Arduino IDE → C/C++ Sketches → Sensors → Actuators → Communication → Projects*

Related: Track 19 — Robotics & IoT | Track 20 — Embedded Systems

Track Overview

Who this is for: Beginners, school/college students, and hobbyists learning microcontroller programming.

Prerequisites: None; ideal first hardware track.

Tools: Arduino Uno/Nano/ESP32, Arduino IDE, breadboard, sensor kit, multimeter, Fritzing.

Hardware kit: Arduino board + jumper wires + resistors + LEDs + basic sensor pack

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	Arduino basics	5 mini projects
1 Month	80–100	Confident builder	Home automation controller
3 Months	250–300	Project-ready	IoT-enabled Arduino capstone
6 Months	500–600	Maker/engineering entry	Portfolio of 8+ projects
9 Months	750–900	Advanced maker	Custom PCB + firmware project

2-Week Crash Course

Days	Topics	Project
D1–2	Electricity basics, breadboard, Ohm's law	LED circuits
D3–4	Digital I/O, analogRead, PWM	Dimming LED + potentiometer
D5–6	Ultrasonic, IR, temperature sensors	Distance + temp monitor
D7–8	LCD display, serial monitor debugging	Sensor dashboard on LCD
D9–10	Servo, buzzer, integration	**Alarm / automation project**

1-Month Foundation

Week	Topics	Project
W1	Arduino IDE, variables, loops, functions	10 basic sketches
W2	Sensors: light, motion, sound	Smart night light
W3	Motors, relays, actuators	Automated plant watering
W4	ESP8266/ESP32 Wi-Fi intro	**Wi-Fi controlled device**

3-Month Job Ready

Month	Topics	Projects
M1	Advanced sensors, interrupts, timers	Weather station
M2	Bluetooth, MQTT, Blynk app control	Mobile-controlled robot car
M3	Multi-module integration, capstone	**Smart home automation system**

6-Month Professional

Month	Topics	Projects
M1–M3	As 3-month track	—
M4	Library development, code organization	Reusable sensor library
M5	PCB design intro (EasyEDA), soldering	Custom shield board
M6	Capstone	**Production-quality Arduino product**

9-Month Advanced

Capstones: IoT product prototype + custom PCB firmware system | **Level:** L5

Assessment Rubric

Component	Weight	Criteria
Weekly Quiz	20%	Electronics, syntax, pin functions
Lab Completion	40%	Circuits work; code uploaded successfully
Project	30%	Creativity, reliability, documentation
Portfolio	10%	Fritzing diagrams, demo videos

Appendix: Mentor Notes

Beginner-friendly entry point for Track 19 and Track 20. Emphasize current-limiting resistors on all LEDs.